

Project 6: Policy-Compliance Verification Agent

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Weekend 5, From Data to Decisions
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An agent which verifies policy compliance

Your company wants to automate the process of handling employee requests

Four types of actions available:

- **Expense Report:** Submit a business expense for reimbursement.
- **Procurement Request:** Request approval to purchase goods or services from a vendor
- **Access / Config Change:** Grant access to a system or change a security configuration
- **Time-off Request:** Request leave / time off

Goal: Build a robust agentic workflow to make sure all actions are compliant with company policy

Company Dashboard

Perform an action. An agent verifies it against company policy, flags problems, cites the rules, and suggests fixes.

Paste your
key here

OpenRouter API key



.....

Kept in this tab only · sent per request, never stored.

Implementation

auto | mode · 0/7 functions live

Finish all TODOs in the notebook
and patch your implementations

7 to go



validate_action

Part 1 · Action



build_context

Part 2 · Context



retrieve_policies

Part 3 · Retrieval



verify_action

Part 4 · Verifier



propose_solution

Part 5 · Solution



run_display_agent

Part 6 · Display



verify

Part 7 · Pipeline

Expense Report

Procurement Request

Access / Config Change

Time-off Request

Expense Report

Submit a business expense for reimbursement.

Category

hotel



Amount (CHF)

Verification

Submit an action to see the agent's verdict.

Grounded on 9 policy documents

Actions & Their Fields

Expense Report

Procurement Request

Access / Config Change

Time-off Request

Action types

Expense Report

Submit a business expense for reimbursement.

Category

taxi

Amount (CHF)

100

Region

switzerland

Receipt attached

☒ Yes

Days since expense

5

Description

ZRH to Zurich HB for customer visit

Submit action →

Verification

Submit an action to see the agent's verdict.

Grounded on 9 policy documents

Fields

- Required / Optional
- Type/Value constraints

Actions violating company policy do not get approved

Expense Report

Submit a business expense for reimbursement.

Category

taxi

△ Taxi expenses are only reimbursable when public transport is impractical (e.g., late at night, heavy equipment, or journey >60 mins). The journey from ZRH to Zurich HB is a standard route well-served by public transport, and no justification for its impracticality was provided.

Amount (CHF)

100

Region

switzerland

Receipt attached

☒ Yes

Days since expense

5

Description

ZRH to Zurich HB for customer visit

Submit action →

Verification

Problematic

The taxi expense violates the policy as the description does not justify why public transport was impractical for a standard route.

MEDIUM category

Taxi expenses are only reimbursable when public transport is impractical (e.g., late at night, heavy equipment, or journey >60 mins). The journey from ZRH to Zurich HB is a standard route well-served by public transport, and no justification for its impracticality was provided.

Suggested fix: Add a justification to the description field explaining why public transport was impractical (e.g., carrying heavy equipment) or change the expense to public transport if no such reason exists.

[expense_reimbursement_policy.md](#)

[Section: Expense Reimbursement Policy > Travel Expenses > Ground Transportation] Taxi and ride-share (Uber, Lyft) expenses are reimbursable when public transport is impractical — for example, late at night, when carrying heavy equipment, or when the journey would exceed 60 minutes by public transport. Employees should choose the economy tier of ride-share services; premium tiers require justification. Rental cars are permitted when a car is genuinely necessary for the work. Employees should choose compact or economy class vehicles. Fuel, tolls, and parking directly related to business travel are reimbursable. Fines and traffic violations are not. Personal car use is reimbursed at CHF 0.70 per kilometre (Switzerland) or the applicable statutory rate in other countries. The starting point is the employee's normal workplace, not their home.

Highlight Issues

severity

verdict

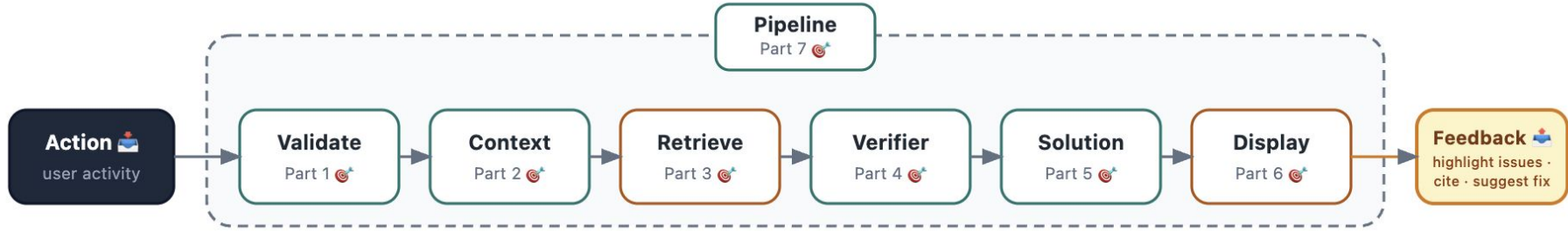
Problematic field

Justification

Proposed solution

Policy source

Controlled workflow gives you robustness



1. Check if action is missing required fields
2. Construct a query for RAG
3. Retrieve most relevant policy chunk for compliance verification
4. Identify problems
5. Propose solutions
6. Flag issues on the dashboard UI
7. Combine all steps into one workflow

Steps

- Complete all TODOs in [WE6_project_student.ipynb](#) in Colab
- Download all the patching files (“Export your functions” section)
- Set up the project branch locally on your laptop: in your terminal run

```
git clone git@github.com:eth-fdd-fs26/FDD-WE6-public.git
```

```
git switch project
```

- Move the solution files you downloaded from Colab into the [solutions/](#) folder
(overwrite the placeholder files)

Steps (continued)

- [Backend] In a terminal, run `uv sync` and then `uv run uvicorn server:app --reload`
- [Frontend] Open another terminal tab at the project folder and run
`cd frontend && npm install && npm run dev`
- Open <http://localhost:5173/> in browser to test the app
- Play with the agent and try to improve

Where to find what

- Data models (input/output schema): [./agentic/models.py](#)
- Prompts (Part 4 & Part 5): [./agentic/prompts.py](#)
- Source policy documents: [./rag/data/](#)
- Example actions for testing: [./examples/](#)
- Solutions files to overwrite: [./solutions/](#)